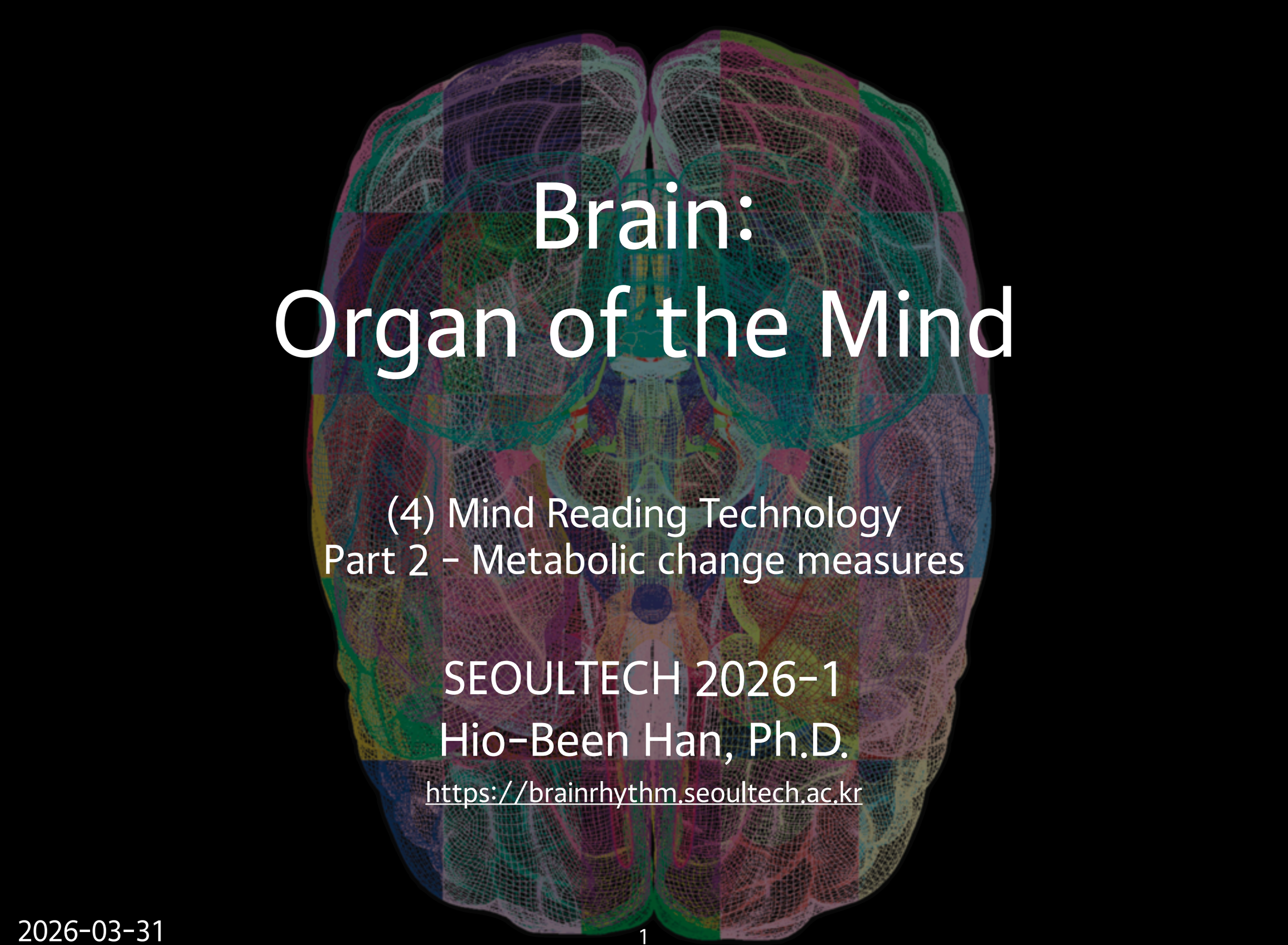


Brain:
Organ of the Mind

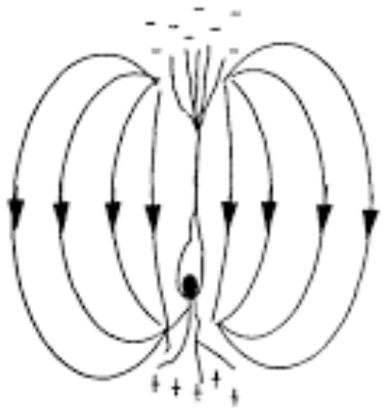
(4) Mind Reading Technology
Part 2 - Metabolic change measures

SEOULTECH 2026-1
Hio-Been Han, Ph.D.
https://brainrhythm.seoultech.ac.kr

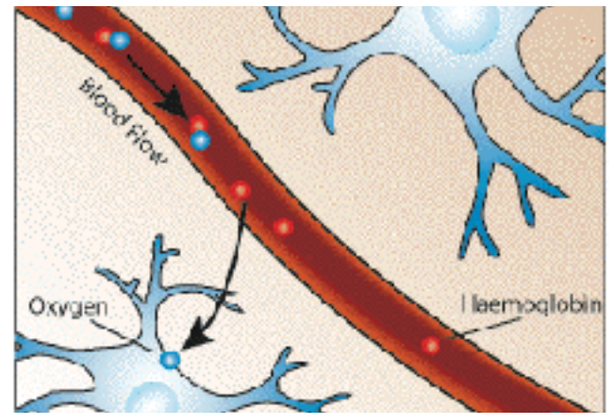
2026-03-31

Two types of functional brain imaging

Electromagnetic field change



Blood flow change



Neural activity

Electric changes (fast)

Metabolic changes (slow)

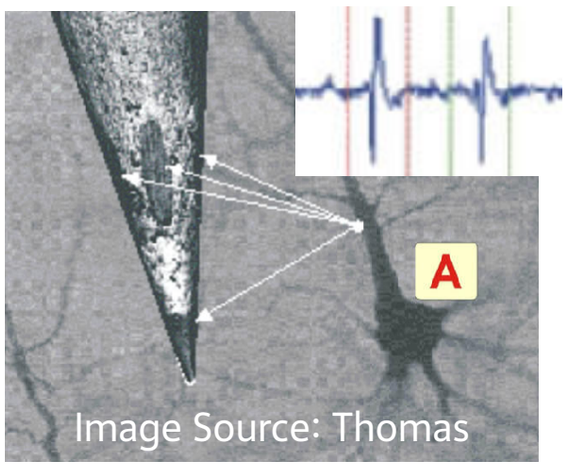
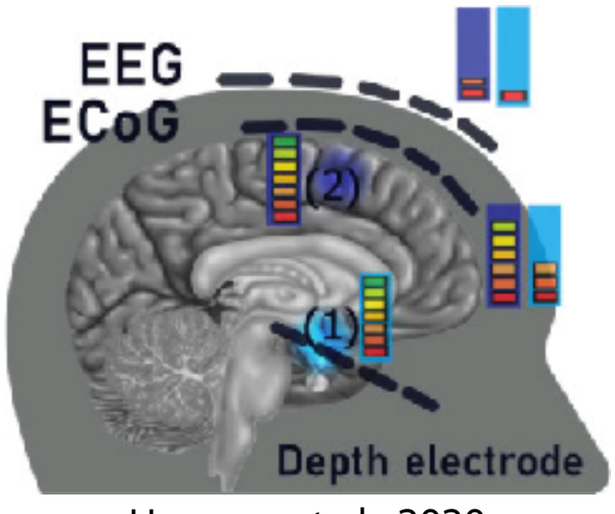


Image Source: Thomas


Image source: NIH

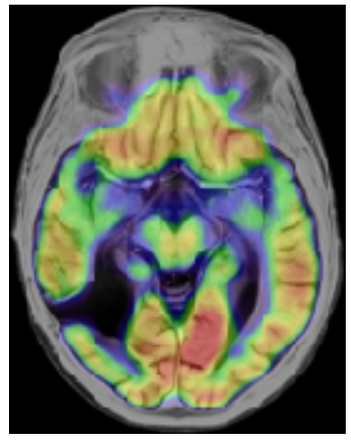
Neural spike (SUA, MUA)	fMRI, PET
EEG, ECoG, LFP, MEG	fNIRs



Hnazaee et al., 2020

High temporal resolution (less than 1 msec)

Relatively low temporal res. (typically 1~2 sec)


Image source: Imaging Tech News

How fMRI innovated brain research



https://www.ted.com/talks/nancy_kanwisher_a_neural_portrait_of_the_human_mind

The first discovery of FFA

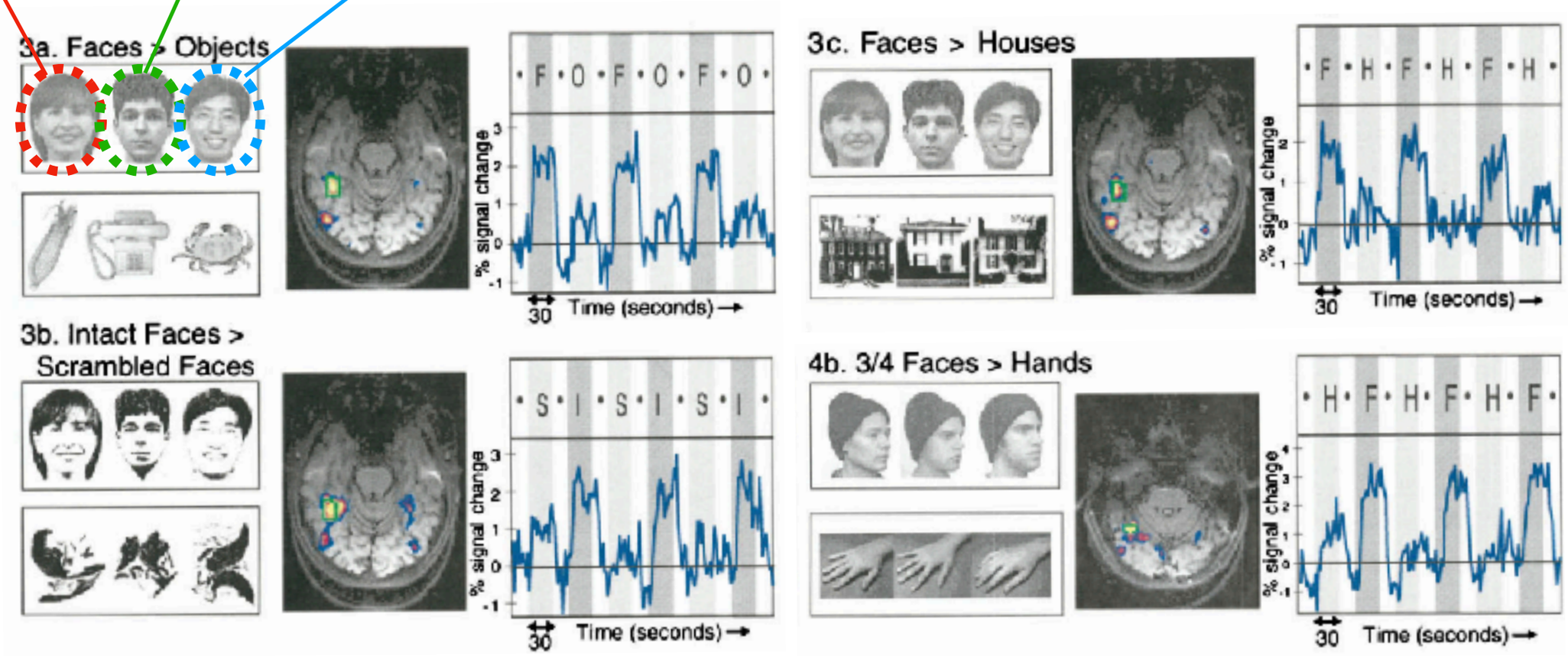
- fMRI measures “increase/decrease” of blood flow within particular brain region, and FFA shows increased blood flow when ‘face’ presents

- The second-most cited paper in the Journal of Neuroscience (more than 10,000 follow-up studies)

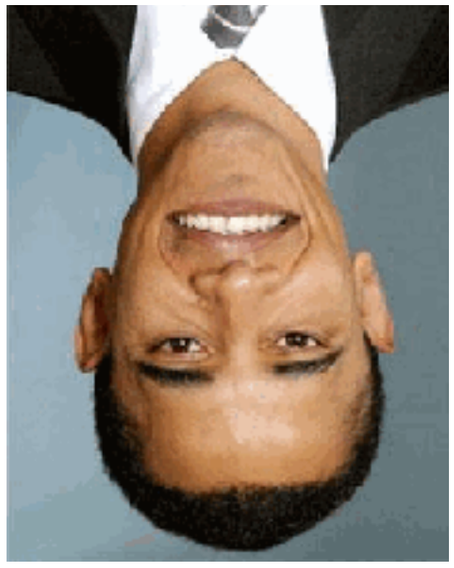
The Fusiform Face Area: A Module in Human Extrastriate Cortex Specialized for Face Perception

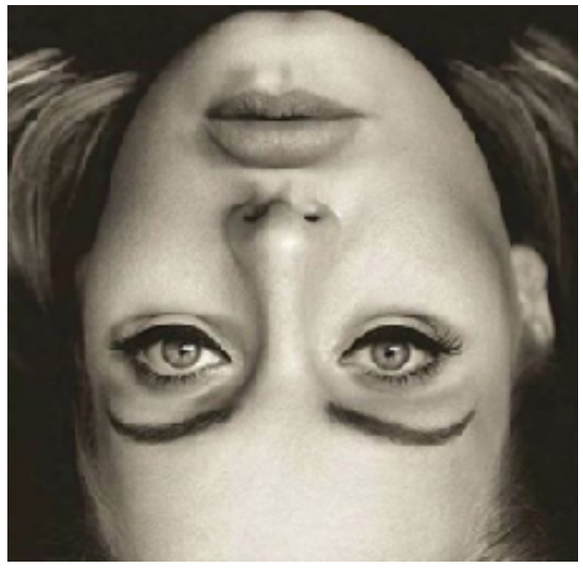
Nancy Kanwisher,[1,2] **Josh McDermott,**[1,2] **and Marvin M. Chun**[2,3]

[1]*Department of Psychology, Harvard University, Cambridge, Massachusetts 02138,* [2]*Massachusetts General Hospital*



Special nature of human face perception





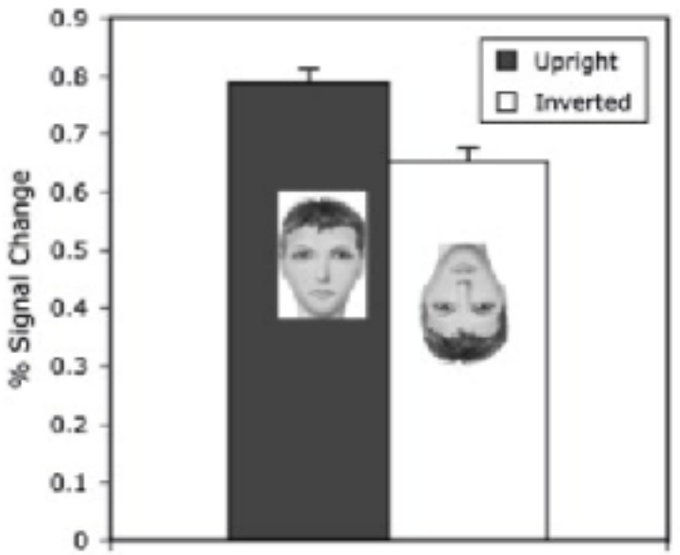


- Human ability for facial perception is very fast and automatic (but heuristic), and specialized brain area (FFA) is in charge of this processing.

- However, when the face is inverted (upside down), this ability suddenly fails. Even though the face image is severely impaired, it's hard to notice (i.e., Thatcher effect).

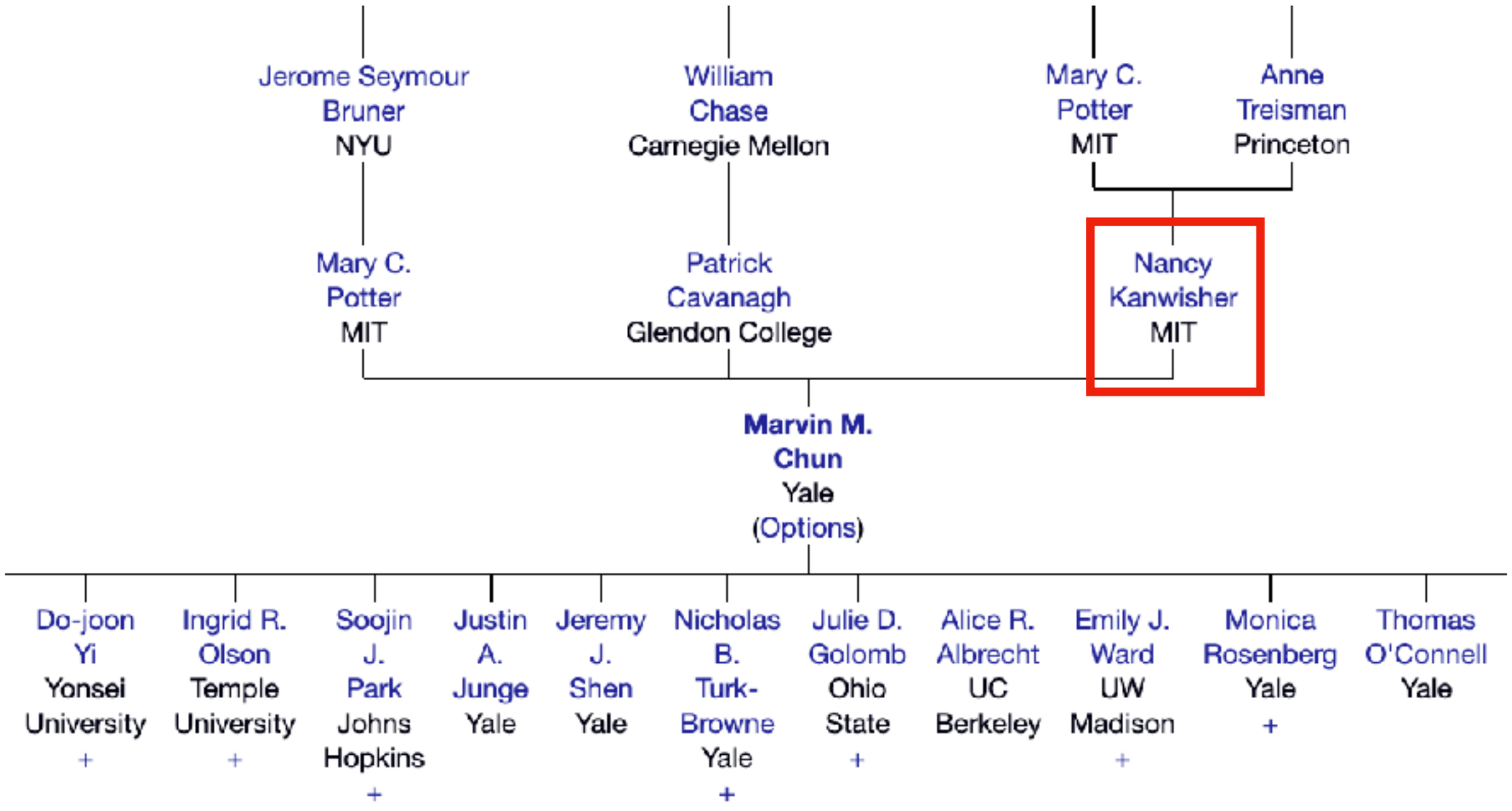
- For inverted face stimuli, FFA activation is decreased compared to upright faces (Yovel & Kanwisher, 2005), reflecting the impaired ability for facial perception.





Academic genealogy (family tree)

https://neurotree.org/neurotree/tree.php?pid=1145



Magical sorting hat?



- The “Sorting Hat” in the Harry Potter movies read a person's mind and extract traits such as intelligence, bravery, and personality.

ARTICLES
https://doi.org/10.1038/s41562-022-01301-1
nature human behaviour
Check for updates

A brain-based general measure of attention

Kwangsun Yoo[1✉], Monica D. Rosenberg[1,2], Young Hye Kwon[1], Qi Lin[1], Emily W. Avery[1], Dustin Sheinost[3], R. Todd Constable[3,4,5] and Marvin M. Chun[1,4,6,7✉]

Attention is central to many aspects of cognition, but there is no singular neural measure of a person's overall attentional functioning across tasks. Here, using original data from 92 participants performing three different attention-demanding tasks during functional magnetic resonance imaging, we constructed a suite of whole-brain models that can predict a profile of multiple attentional components (sustained attention, divided attention and tracking, and working memory capacity) for novel individuals. Multiple brain regions across the salience, subcortical and frontoparietal networks drove accurate predictions, supporting a common (general) attention factor across tasks, distinguished from task-specific ones. Furthermore, connectome-to-connectome transformation modelling generated an individual's task-related connectomes from rest functional magnetic resonance imaging, substantially improving predictive power. Finally, combining the connectome transformation and general attention factor, we built a standardized measure that shows superior generalization across four independent datasets (total $N = 495$) of various attentional measures, suggesting broad utility for research and clinical applications.

Resting state brain scan → Attention task performance

10 minutes of brain scanning can predict other test scores by up to 20% of variance
(= It can replace IQ test, at some extent!)

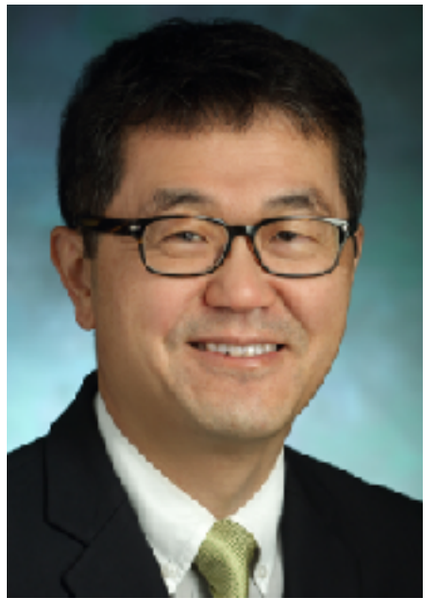
fMRI application in business



Heungyeol Lee

neurogazer CEO

Co-founder



Daeyeol Lee

Prof. at Johns Hopkins Univ

Co-founder





Marvin Chun

Prof. at Yale Univ

Advisor/Collaborator

(f)MRI data

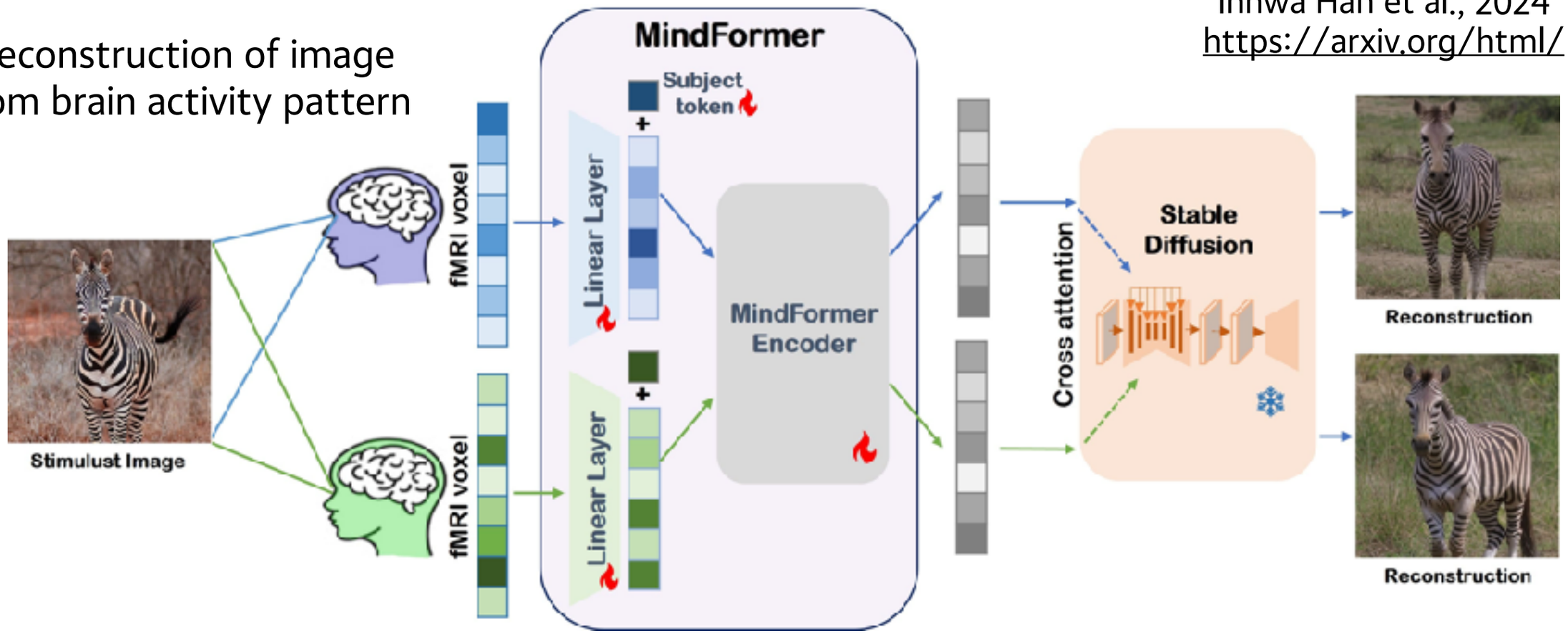


Predict child's IQ, artistic ability, mathematical skills, foreign language proficiency, etc.

fMRI application boosted by AI tech

Inhwa Han et al., 2024
https://arxiv.org/html/

Reconstruction of image
from brain activity pattern





Beginning of medical imaging: X-ray (2D)

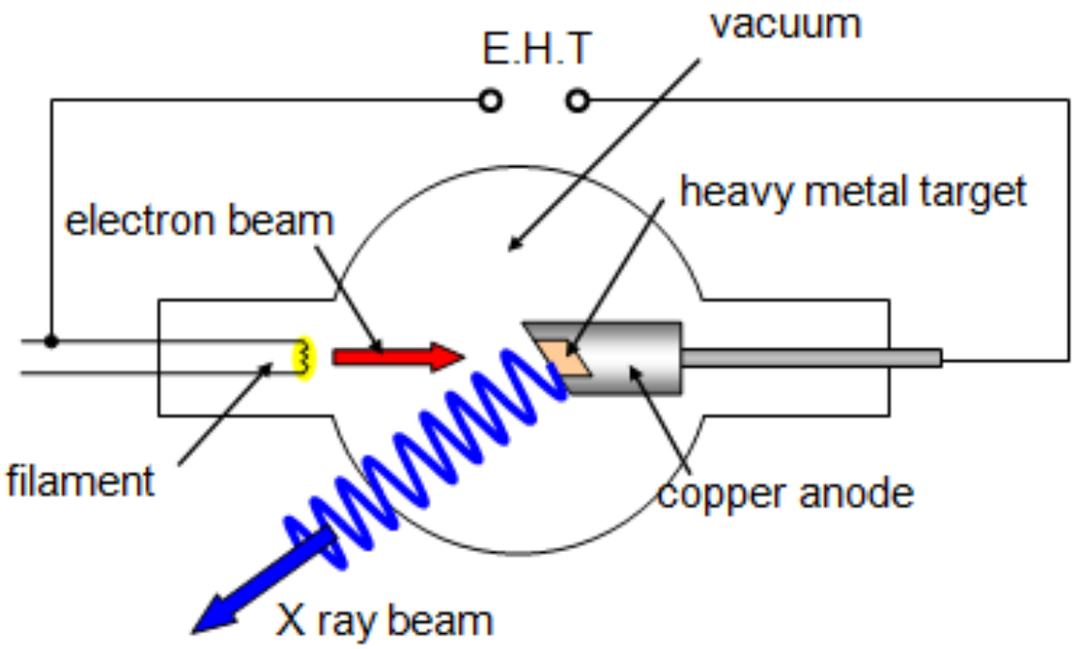
- **Wilhelm Röntgen, German physicist, produced and detected electromagnetic radiation in a wavelength range known as X-ray (radioactive).**

- **Röntgen stumbled upon X-rays while working on other thing (cathode rays), but later became first winner of the Nobel Prize in Physics (1901).**

- **Despite the potential for profit, he chose to share his discovery with humanity (no patent, for free!), significantly advancing science and technology—including contributions to fields like DNA research.**







Non-medical application of X-ray

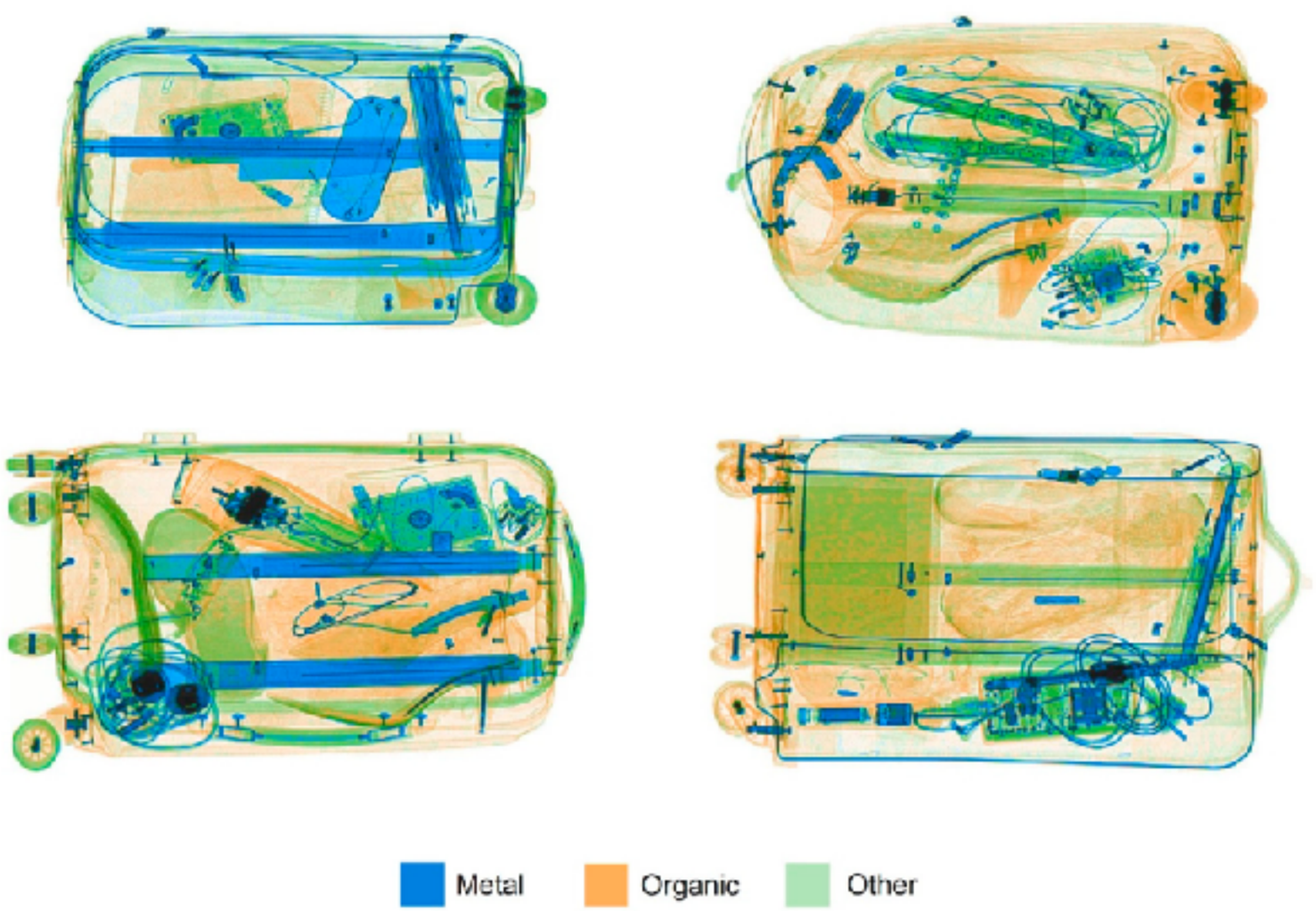




Image source: CaSePIX image library

Image source: Ocala.com

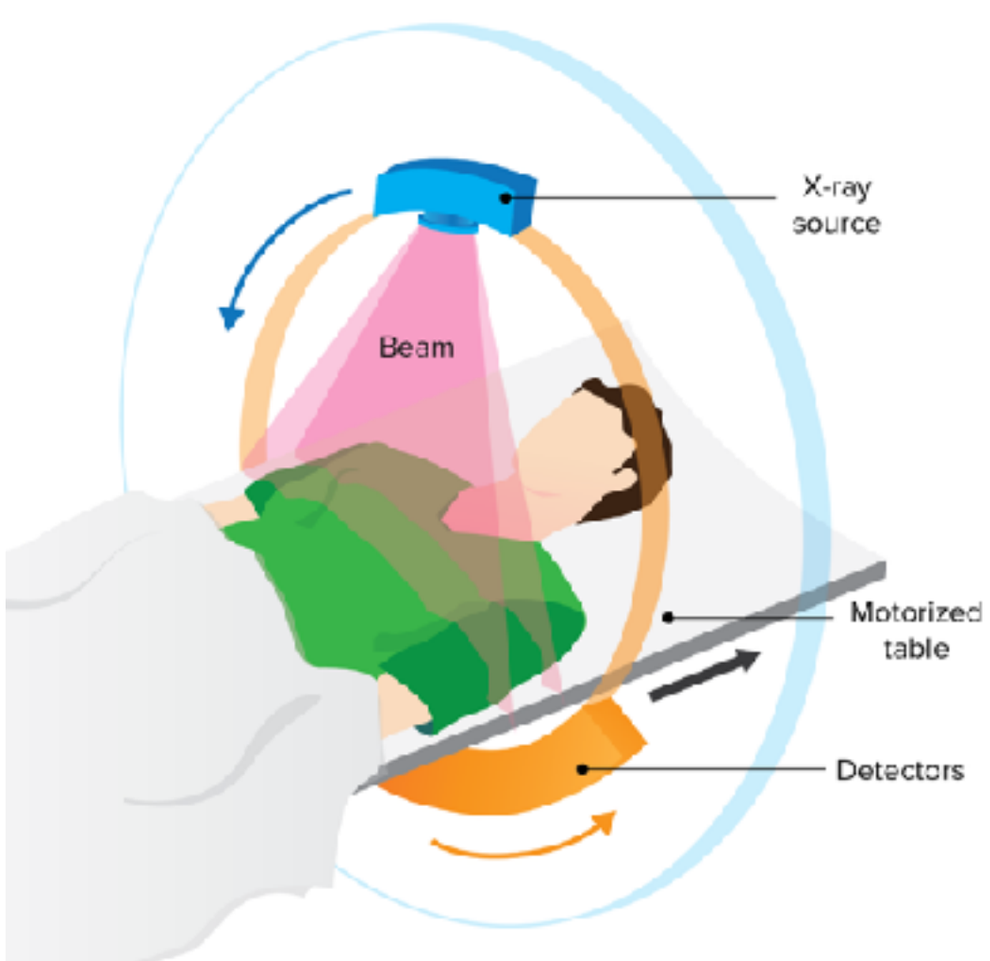
Medical application of X-ray imaging

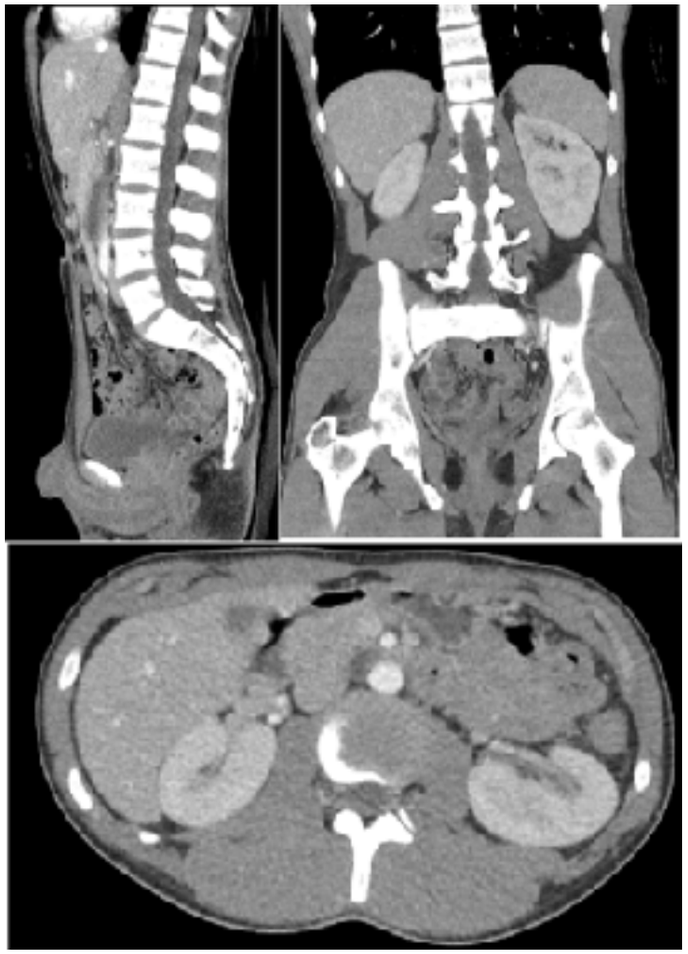


https://youtu.be/nLzPsYyVD6w?si=Xt3rRKIdi7P9QRfu

Structural imaging: CT scan (3D)

- **Computerized Tomography (CT): Constructing 3D image by repeating multiple X-ray imaging (Nobel Prize in Physiology & Medicine, 1979).**
- **Because X-ray is radioactive, CT makes a person be exposed to radiation.**
- **Simple procedures (fast scanning time, ~10 minutes), very high contrast for hard tissues such as bones.**





Structural imaging: MRI (3D)

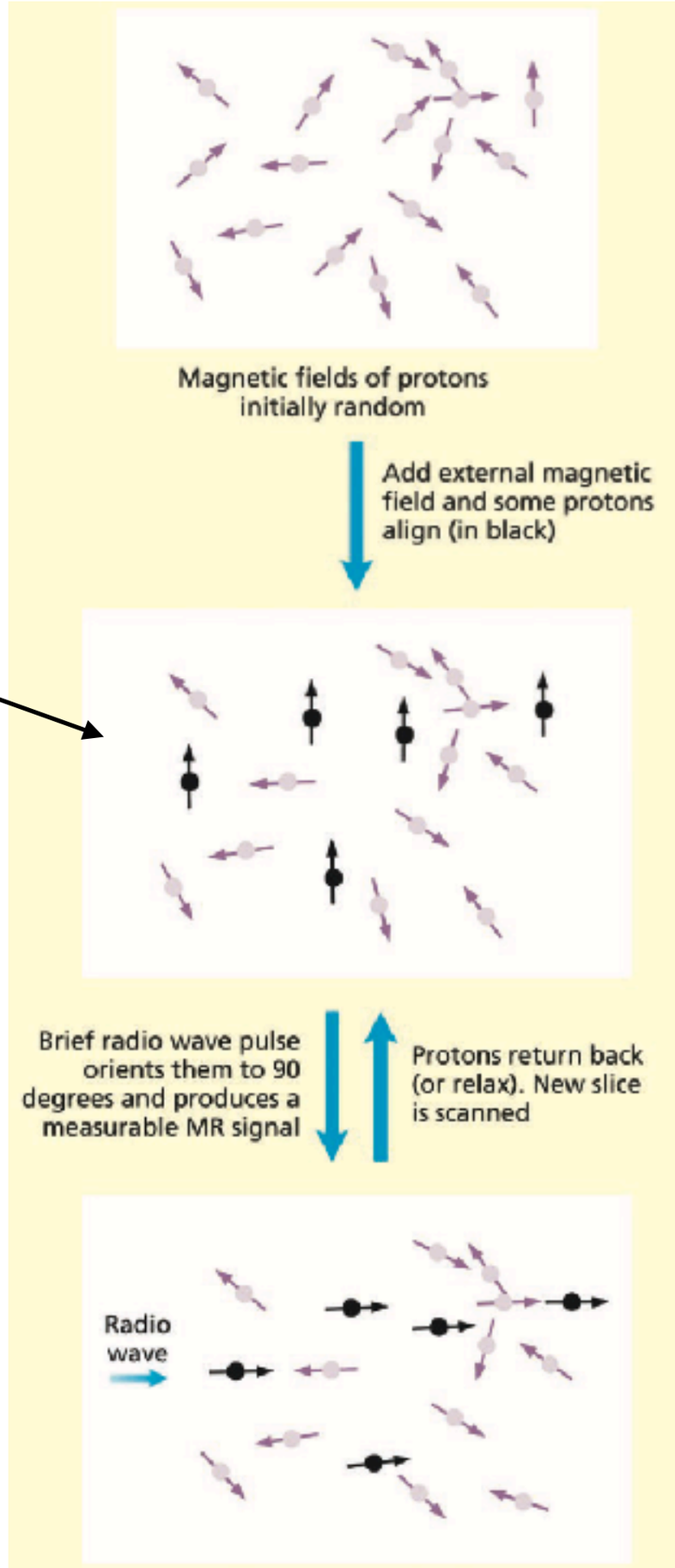
- Magnetic Resonance Imaging (MRI) uses strong magnet to produce NMR phenomena.

- NMR (Nuclear Magnetic Resonance): A property of atomic nuclei (mostly Hydrogen) absorbing and re-emitting radio waves at specific frequencies, when placed in a strong magnetic field (Bloch & Purcell, 1952 Nobel Prize in Physics).

- Later, Paul Lauterbur (physicist) & Peter Mansfield (chemist) developed MRI in 1970s, and won the Nobel Prize in Physiology/Medicine (2003).

- MRI uses signals emitted by Hydrogen after magnetic pulse. Since the hydrogen atoms proportion vary across tissues (skin, muscle, etc.), this allows the differentiation of various tissues for 3D imaging.

- It outperforms CT for soft tissues (such as brain), but takes more time for imaging (typically takes 1 hour for one person scan).



Most complicated math in brain science

- Angular momentum

$$\vec{\mu} = \gamma \vec{J}$$

Gyromagnetic ratio
(2.π rad *42.58Mhz/T for ^{1}H)

$$\gamma = \frac{\mu}{J} = \frac{q\pi r^2/T}{m2\pi r^2/T} = \frac{q}{m}$$

Energy Difference, Bulk Magnetization

$$\Delta E = h\nu = \hbar \frac{\gamma}{2\pi} B_0$$ Larmor frequency

$$P_p + P_a = 1$$

$$\frac{P_p}{P_a} = \exp(\Delta E/k_B T) \simeq 1 + \frac{\Delta E}{k_B T}$$

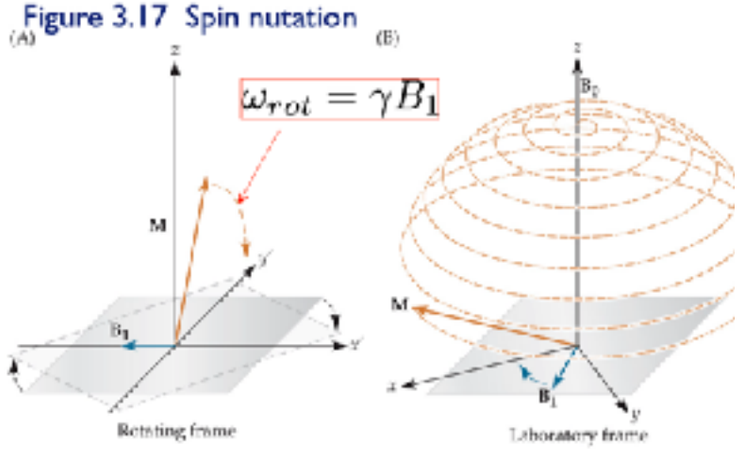
$$M_z = (P_p - P_a) n \mu_z = \frac{\Delta E}{2k_B T} n \mu_z$$

Rotating Frame of Reference

- Clockwise rotating at an angular frequency ω
- Larmor-rotating frame: $\omega = \omega_0$
- RF-rotating frame: $\omega = \omega_{RF}$

$$\begin{aligned} \vec{i'} &= \cos(\omega t)\vec{i} - \sin(\omega t)\vec{j} \\ \vec{j'} &= \sin(\omega t)\vec{i} + \cos(\omega t)\vec{j} \\ \vec{k'} &= \vec{k} \end{aligned}$$

Figure 3.17 Spin nutation



$$\omega_{rot} = \gamma B_1$$

The Block Equation

- Laboratory frame

$$\frac{d\vec{M}}{dt} = \gamma \vec{M} \times \vec{B} - \frac{M_x \vec{i} + M_y \vec{j}}{T_2} - \frac{(M_z - M_z^0)\vec{k}}{T_1}$$

- Rotating frame

$$\begin{aligned} \frac{\partial \vec{M}_{rot}}{\partial t} &= \gamma \vec{M}_{rot} \times \vec{B}_{eff} - \frac{M_{x'}\vec{i'} + M_{y'}\vec{j'}}{T_2} - \frac{(M_{z'} - M_z^0)}{T_1} \\ \vec{B}_{eff} &= \vec{B}_{rot} + \frac{\omega}{\gamma} \end{aligned}$$

Excitation of a Spin System

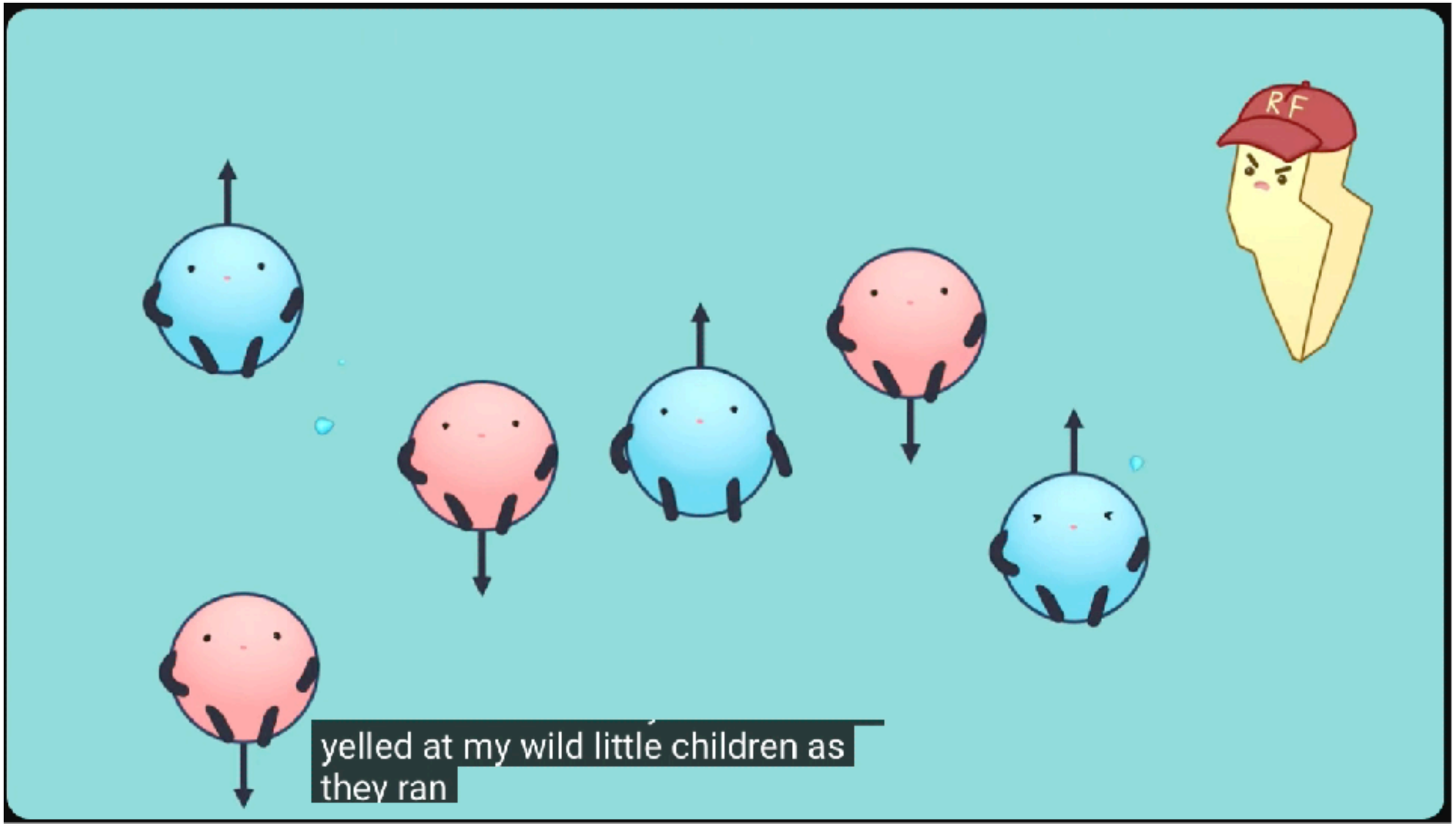
$$\frac{d\mathbf{M}}{dt} = \gamma \mathbf{M} \times \mathbf{B}$$

$$\frac{\delta \mathbf{M}_{rot}}{\delta t} = \frac{\delta \mathbf{M}}{\delta t} - \omega \times \mathbf{M}_{rot} = \gamma \mathbf{M}_{rot} \times \mathbf{B}_{eff}$$

$$\mathbf{B}_{eff} = \mathbf{z}\left(B_0 - \frac{\omega}{\gamma}\right) + \mathbf{x}' B_1$$

This becomes zero at resonance frequency

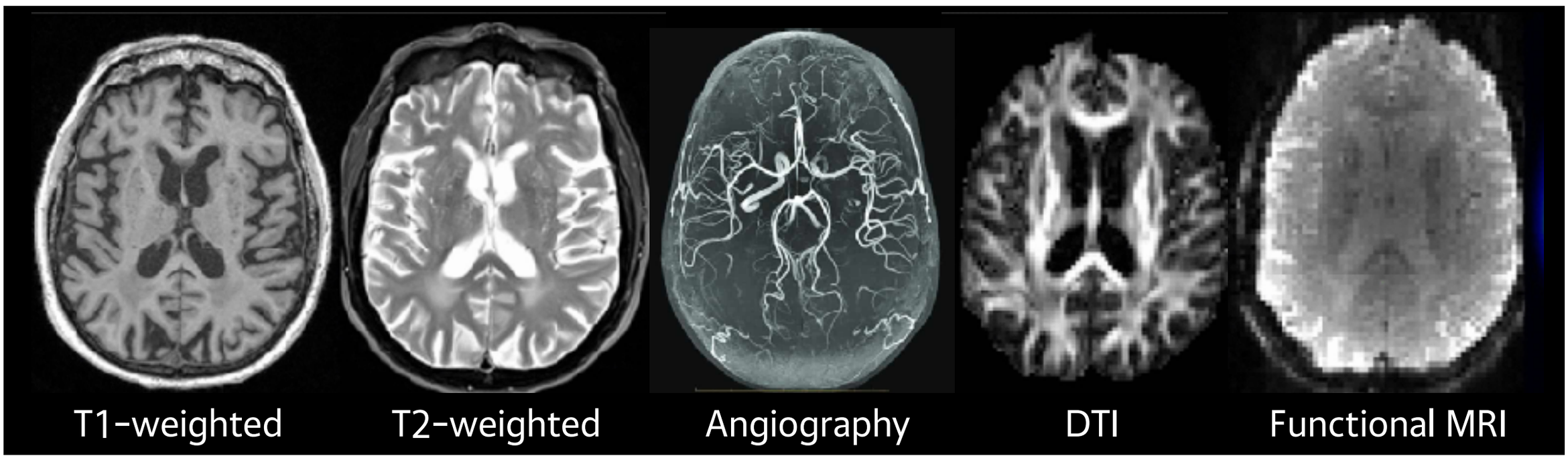
MRI: Magnetic Resonance Imaging



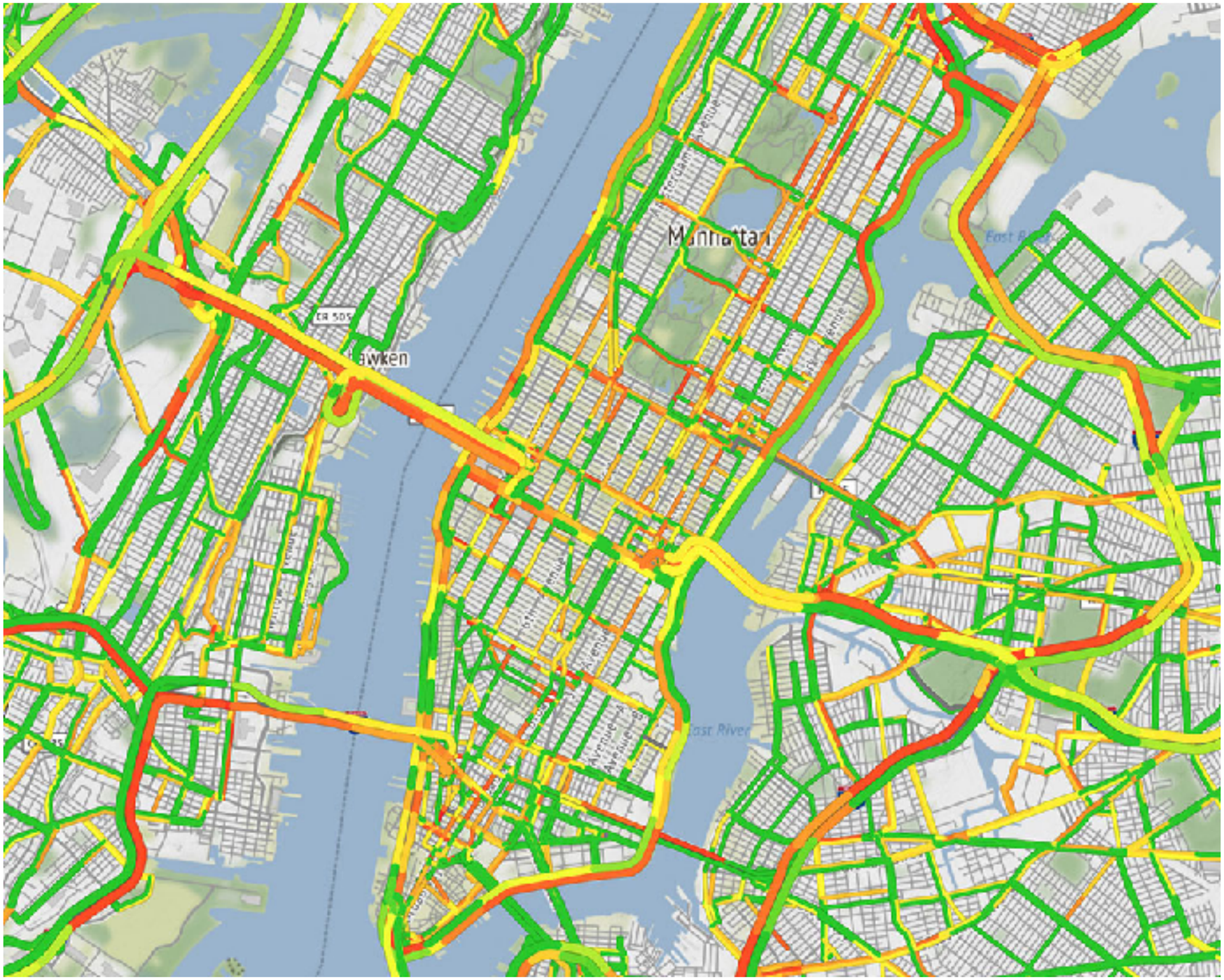
https://youtu.be/jLnuPKhKXVM?si=KzpB5TMffQLaiPI8

Variations of MR imaging

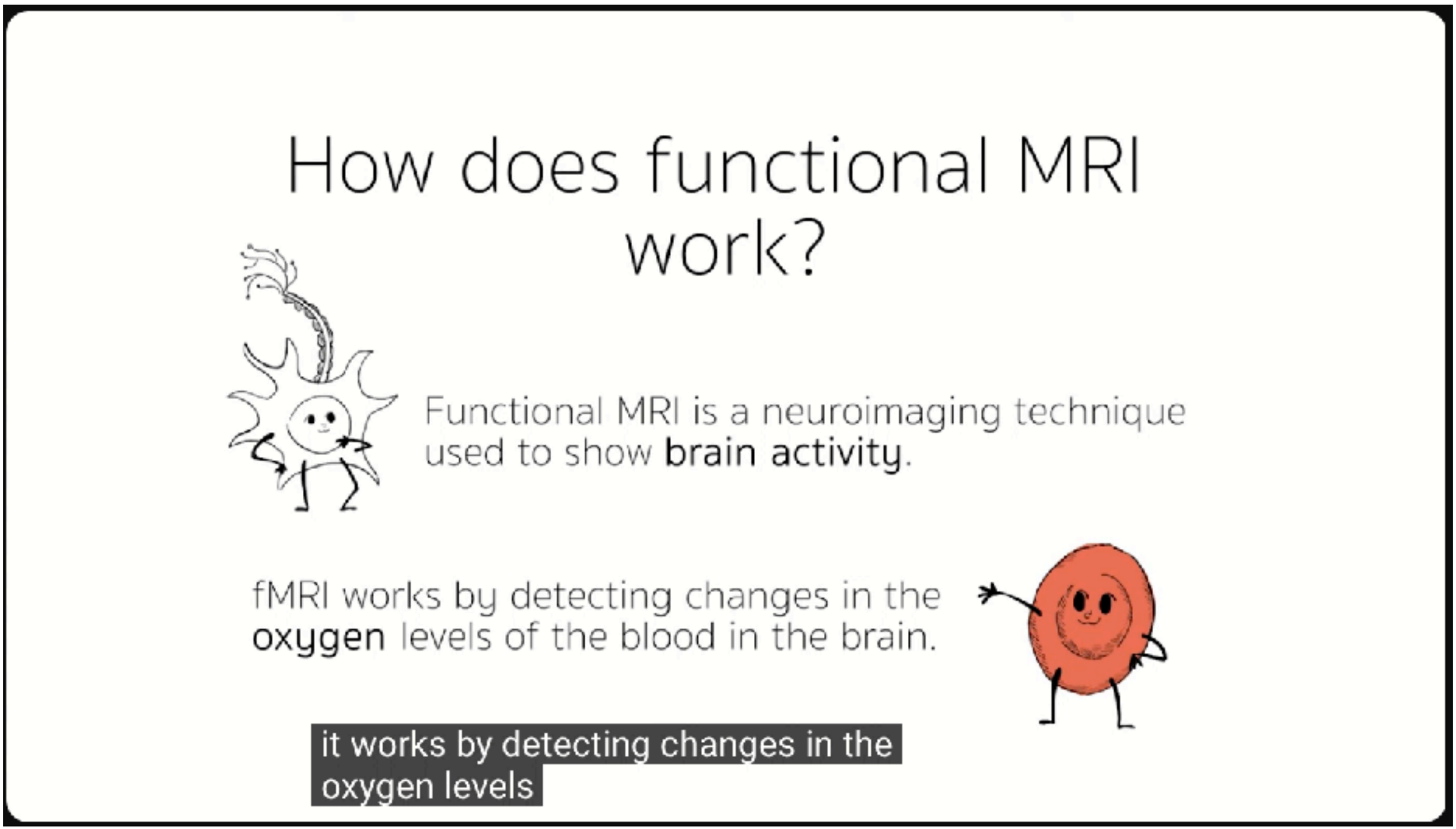
- Design of imaging sequence allow various imaging results.
  - T1 image: Highlights fat and anatomical detail. Best for normal brain anatomy.
  - T2 image: Highlights watery area (e.g., CSF, edema) more in detail.
  - Angiography: Captures blood vessels (based on magnetic property of blood).
  - Diffusion Tensor Imaging (DTI): Captures white matter tracts (based on water molecules' movement)
  - Functional MRI (fMRI): Captures blood oxygenation. Rapidly scans multiple times to observe changes, primarily focused on brain activity mapping.



Anatomical vs. functional imaging

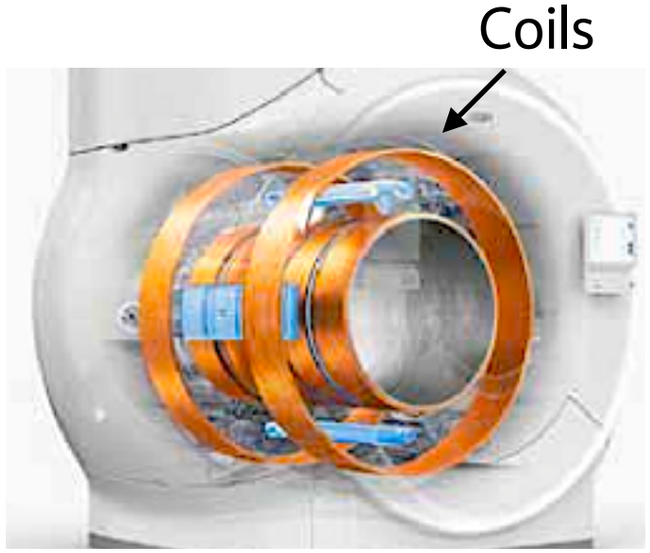


fMRI: functional MRI



https://www.youtube.com/watch?v=jLnuPKhKXVM

Impact of strong magnet (magnetic field)

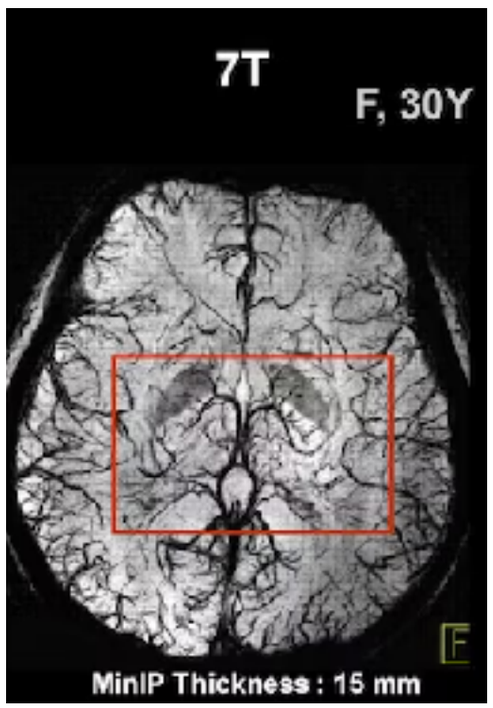


- **Quality of NMR signal depends on the power of magnet (The stronger magnet, the better signal).**

  e.g.) 1.5T (tesla) -> 3T -> 7T -> 11.7T

- **Strong magnet can be dangerous sometimes.**





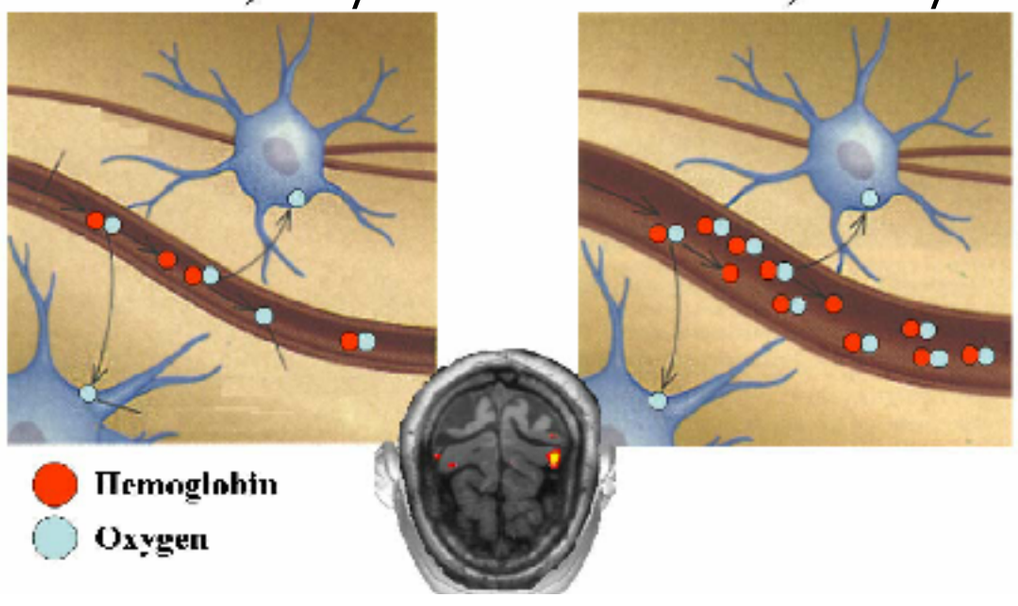


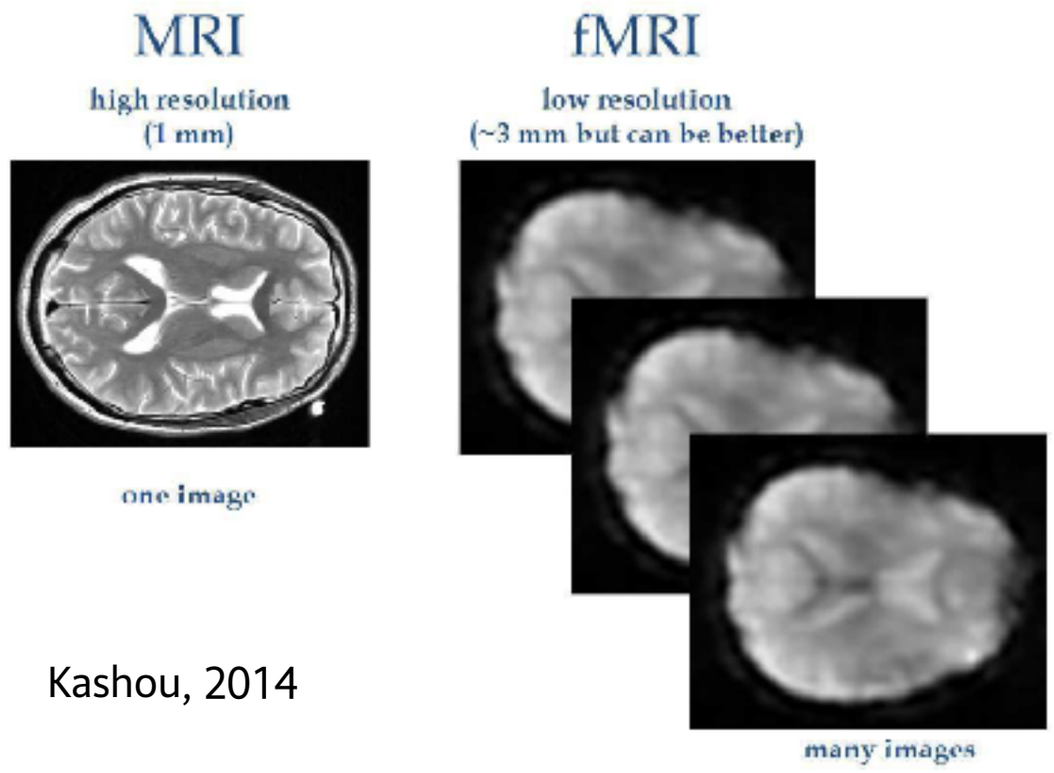
https://www.youtube.com/watch?v=6BBx8BwLhqg

Image credit: Melbourne Univ

fMRI measures BOLD signal

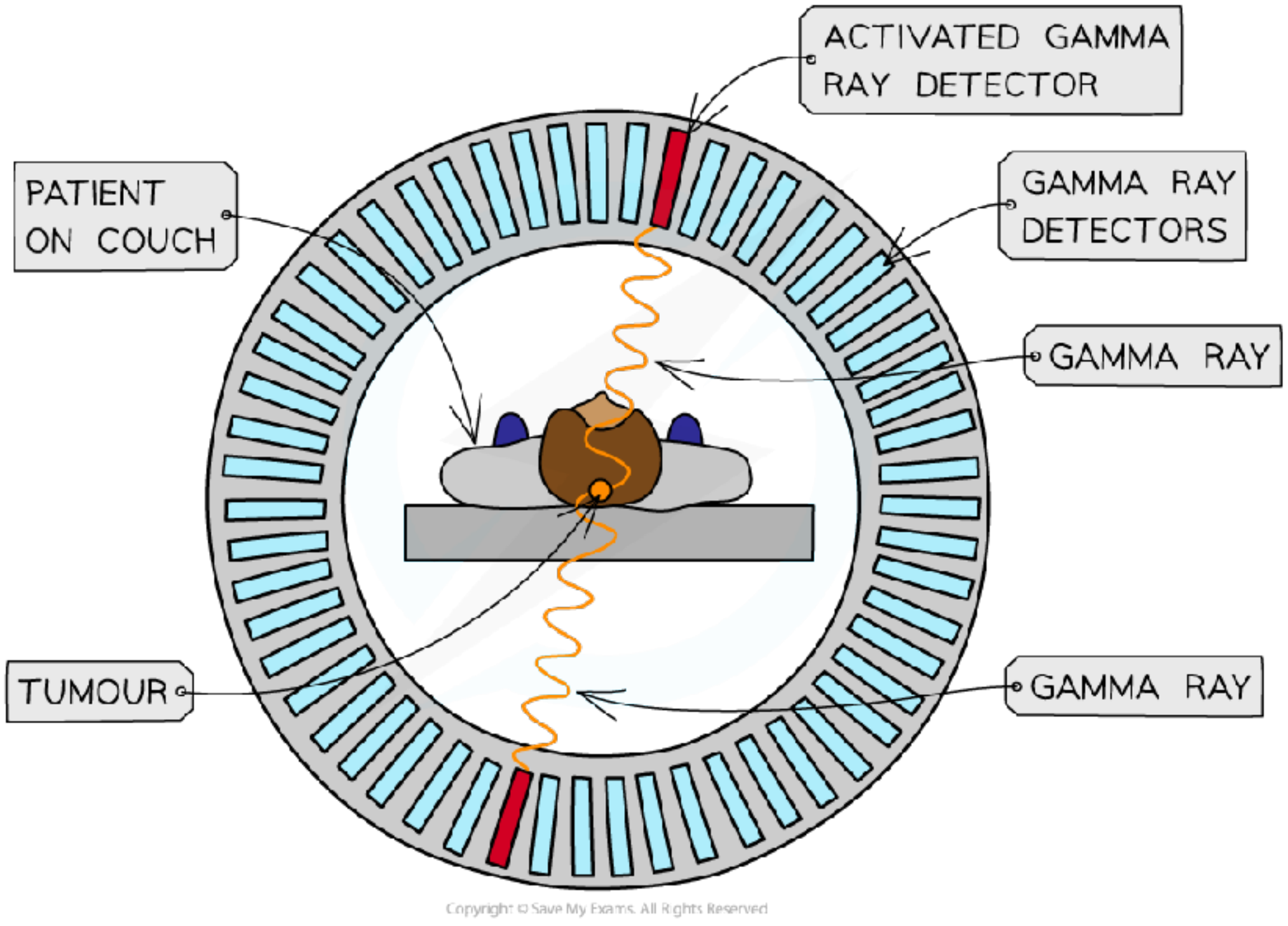
- Blood-Oxygen-Level-Dependent (BOLD) signal from hemodynamic activity
  - Developed by Seiji Ogawa in 1990, showing that blood oxygen changes could be detected via MRI, leading to the development of fMRI.
  - Principle: Magnetic properties of hemoglobin — oxygenated hemoglobin is diamagnetic (repelled by magnets), while deoxygenated hemoglobin is paramagnetic (attracted by magnets). When brain regions become active, they consume more oxygen, and the resulting increase in blood flow alters the MRI signal, generating the BOLD contrast.
- Because it requires rapid imaging (e.g., every 1–2 seconds), spatial resolution is relatively lower.





Kashou, 2014

PET: Radioactive tracer & gamma ray



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Image source: SaveMyExam

PET imaging for metabolic changes

- **Positron Emission Tomography**
  - **Principle: PET uses radioactive tracers injected into the body (in vessels), which emit positrons (and gamma-rays) as they decay, which are detected by scanners.**
  - **Functionality: Stronger activation for active body (brain) region. PET measures metabolic activity by tracking the distribution of tracers, linked to blood flow and oxygen/glucose metabolism. This helps visualize areas with increased energy consumption.**
  - **Temporal resolution: Poor, because it depends on tracer decaying time.**
  - **Despite its limitations, PET is clinically useful because it can track many biochemical processes (which fMRI can't do), especially in fields like cancer and dementia research.**



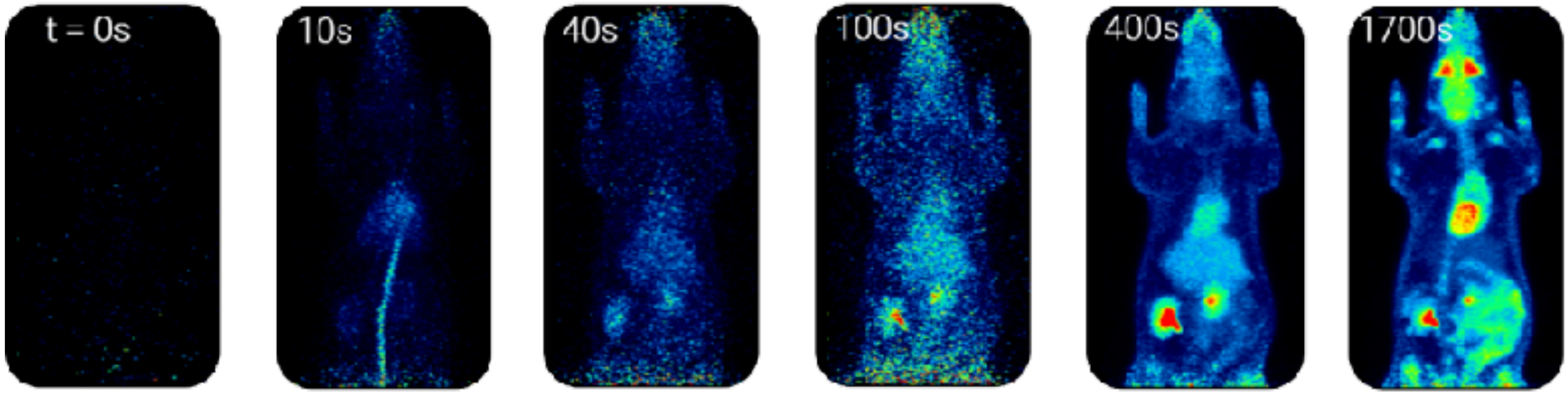


Image source: Bruker

Variations of PET imaging

- Tracer type (injected substance) determines the result of PET.
  - FDG PET: FDG is a substance similar to glucose. When injected into the body, it accumulates in areas with high glucose metabolism. Since cancer cells exhibits high glucose metabolism, FDG can detect activity driven by cancer cells.
  - Amyloid/Tau PET: Amyloid and Tau are key proteins linked to Alzheimer's disease. In Amyloid/Tau PET, a small amount of a substance that binds to Amyloid or Tau is injected, which allows PET to detect the extent of toxic protein deposits in the brain.

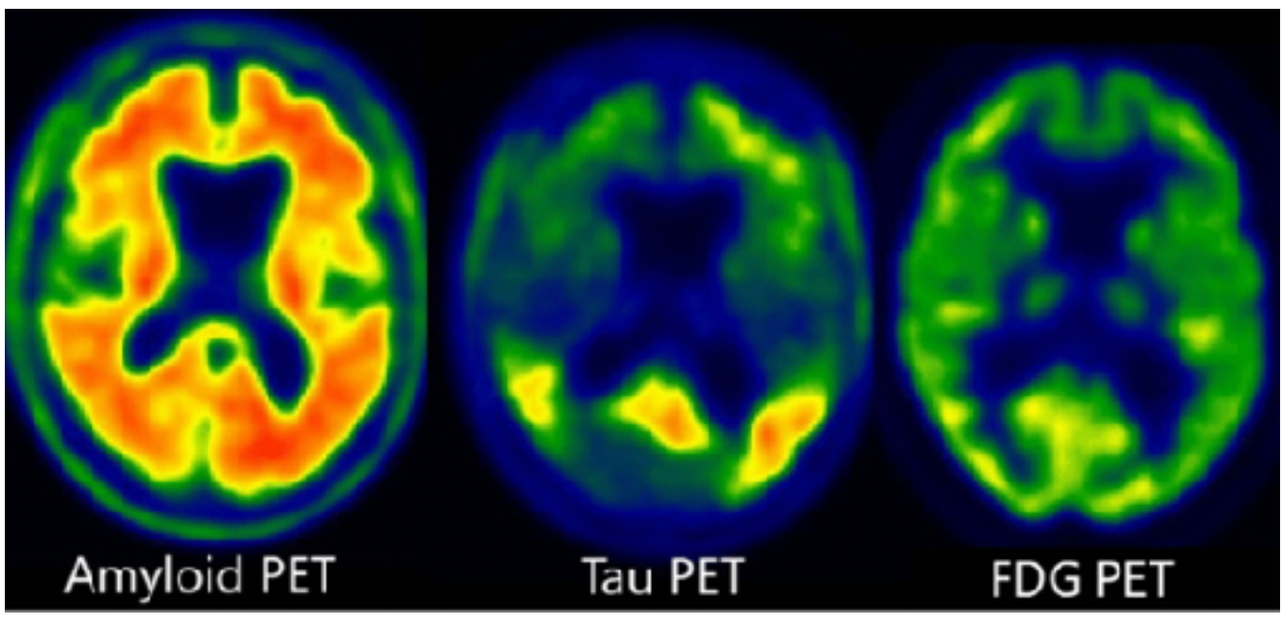


Image source: Schwarz, 2021

Amyloid PET result

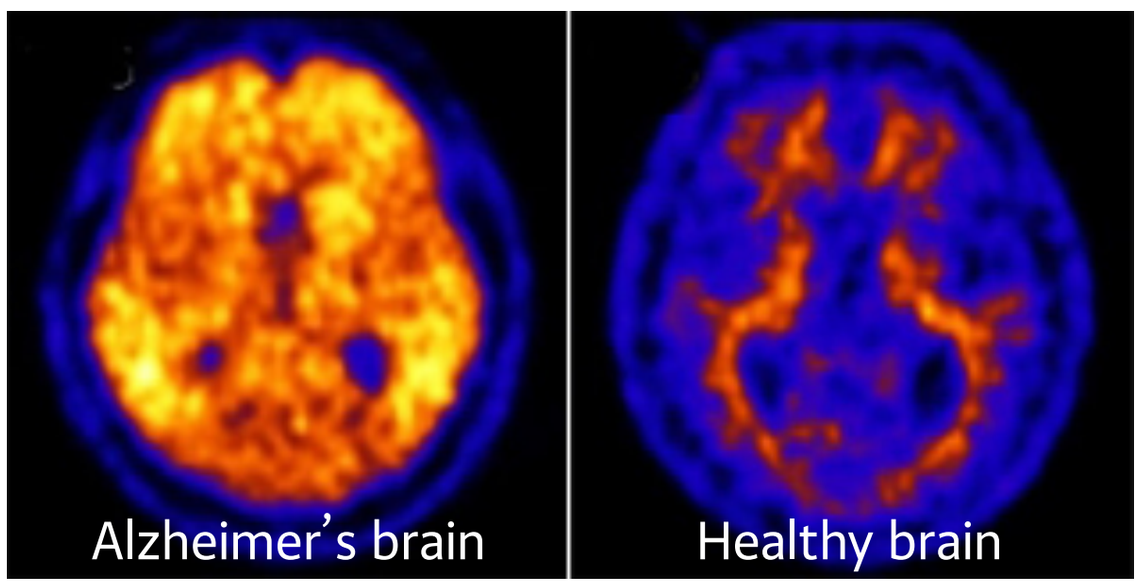


Image source: SNUH

functional Near-Infra-Red Spectroscopy

- **Principle: fNIRS uses light (NIR) to measure changes in oxy/deoxy hemoglobin in the brain. The light penetrates the head, and its absorption varies depending on the oxygenation levels of the blood, indicating metabolic changes after neural activity.**
- **Resolution: Spatial coverage is limited to surface area. Temporal resolution is similar to fMRI (both depends on deoxy/oxy hemoglobin)**
- **Temporal resolution is better than PET but similar to fMRI, allowing real-time monitoring of brain activity. No need for big scanner facility.**
- **Application: fNIRS is commonly used for functional brain activity monitoring, especially in cognitive function studies, where its portability and non-invasiveness are key.**

Vesoulis et al., 2021

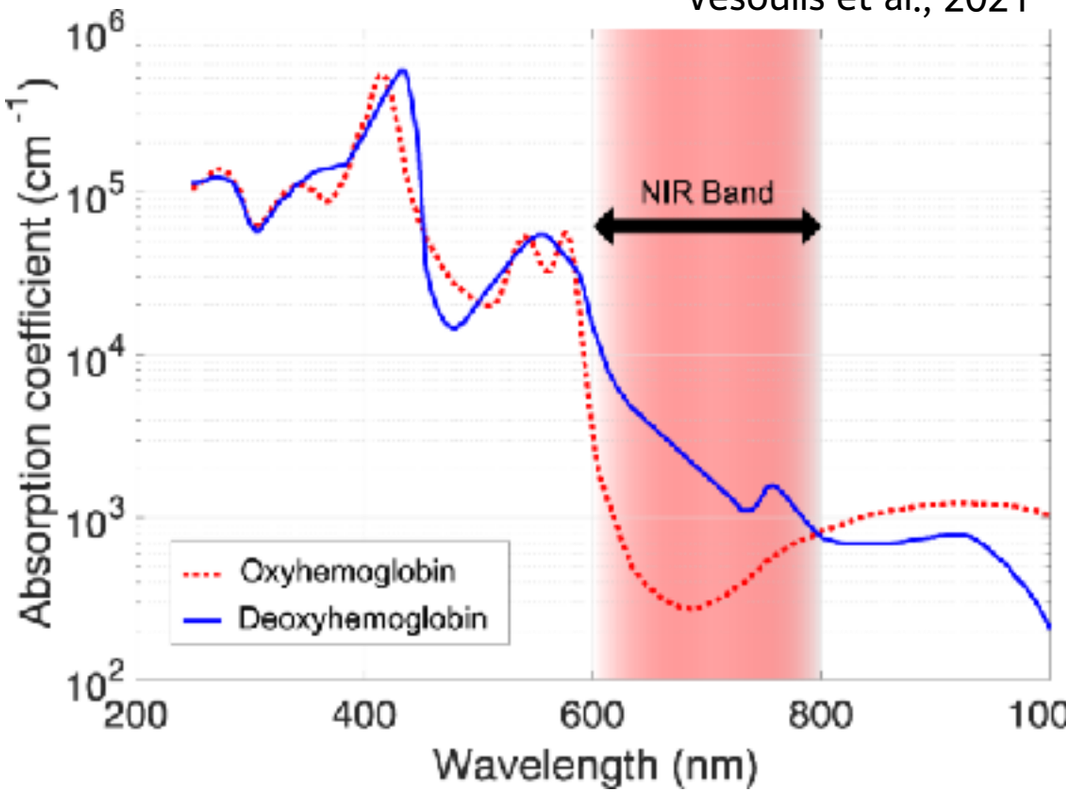




Image source: Cortech Solutions

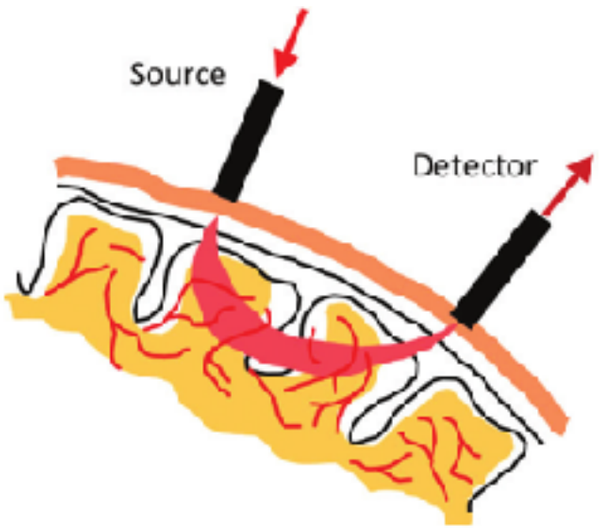


Image source: NIRx Medical

Quick summary

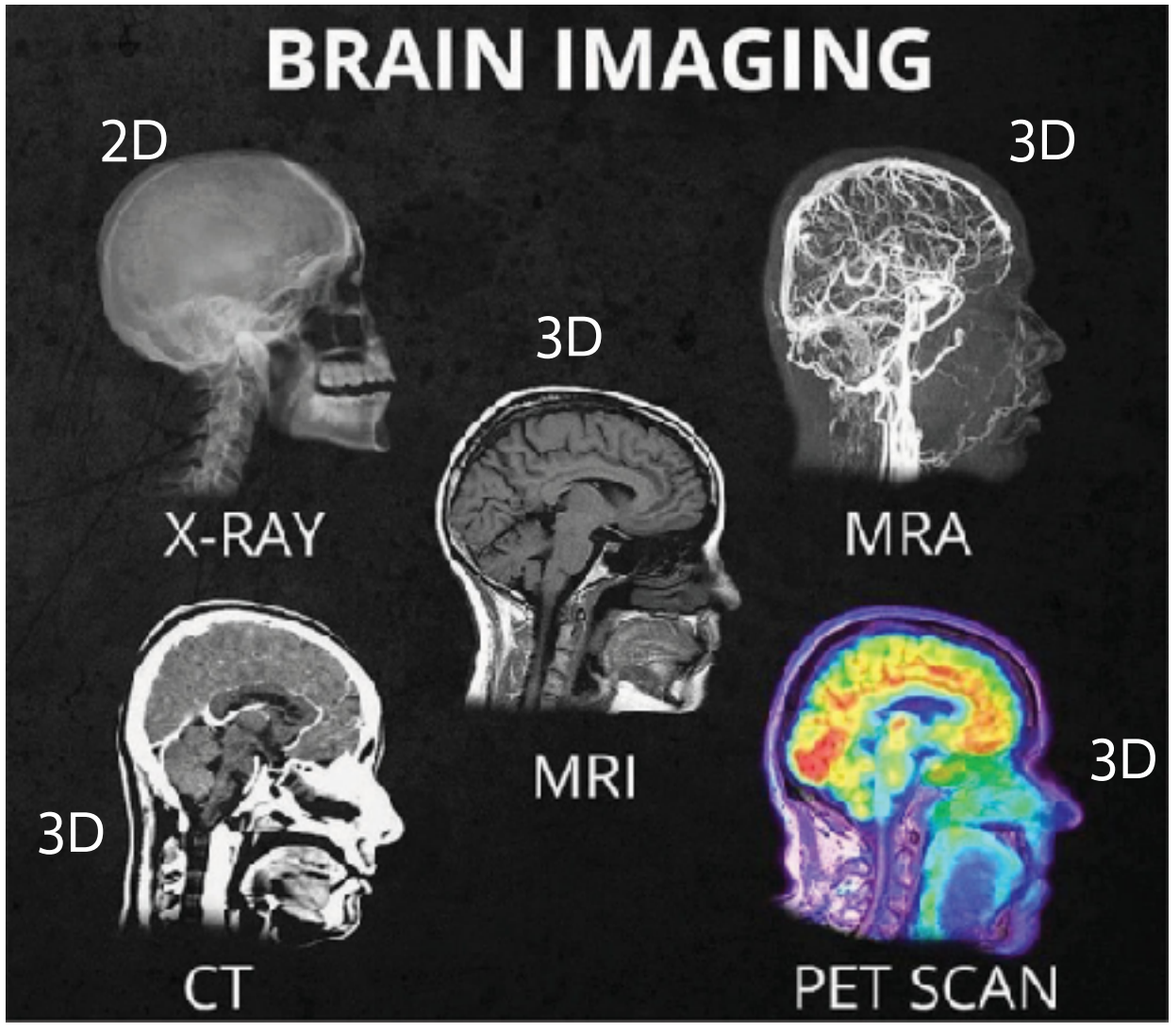


https://www.youtube.com/watch?v=N2apCx1rllQ

Quick summary

- **Structural imaging methods**
  - **X-ray: Uses radioactive wave (fast & cheap)**
  - **CT: Uses radioactive wave (multiple X-ray scanning)**
  - **MRI: Uses NMR (proportion of hydrogen atoms, slow scan)**

- **Functional imaging methods**
  - **fMRI: Uses NMR (change of oxygen level, fast scan)**
  - **PET: Uses radioactive tracer (change of oxygen/glucose level)**
  - **fNIRS: Uses NIR (change of oxygen level, portable)**



https://www.youtube.com/watch?v=N2apCx1rlIQ

Comparison of imaging methods

	Imaging Type	Core principle	Strengths	Weaknesses
X-ray	Structural	Uses **X-rays** (radioactive rays) to create images based on the absorption by different tissues	- Fast (~1 minute) and cheap - Widely available	- Involves radiation exposure - Limited soft tissue contrast
CT	Structural	Uses **X-ray beams** and computer processing to generate detailed cross-sectional images	- Detailed 3D images of hard tissues (such as bones) - Fast (~10 min) for emergency situations	- Radiation exposure - Less effective for soft tissue contrast compared to MRI
MRI	Structural	Utilizes **Nuclear Magnetic Resonance (NMR)**, detecting changes in hydrogen atoms in a magnetic field	- Excellent for soft tissues - No radiation exposure	- Expensive, slow (~60 min) - Not suitable for patients with metal implants
fMRI	Functional	Measures changes in **blood oxygenation** levels (BOLD signal), which are correlated with neural activity	- Good spatial resolution (than PET/ fNIRs imaging) - Allows for both anatomical and functional data	- Expensive, time-consuming - Requires the subject to stay still for long periods
PET	Functional	Uses **radioactive tracers** injected into the body to measure metabolic activity	- Good for detecting metabolic activity - Useful in cancer and brain function studies	- Involves radiation exposure - Lower spatial resolution than fMRI
fNIRS	Functional	Uses **near-infrared light** to measure changes in oxygenated and deoxygenated hemoglobin	- Very cheap - Portable and affordable - Suitable for natural settings (no need of large scanner)	- Limited depth penetration - Lower spatial resolution than fMRI

Comparison of functional imaging methods

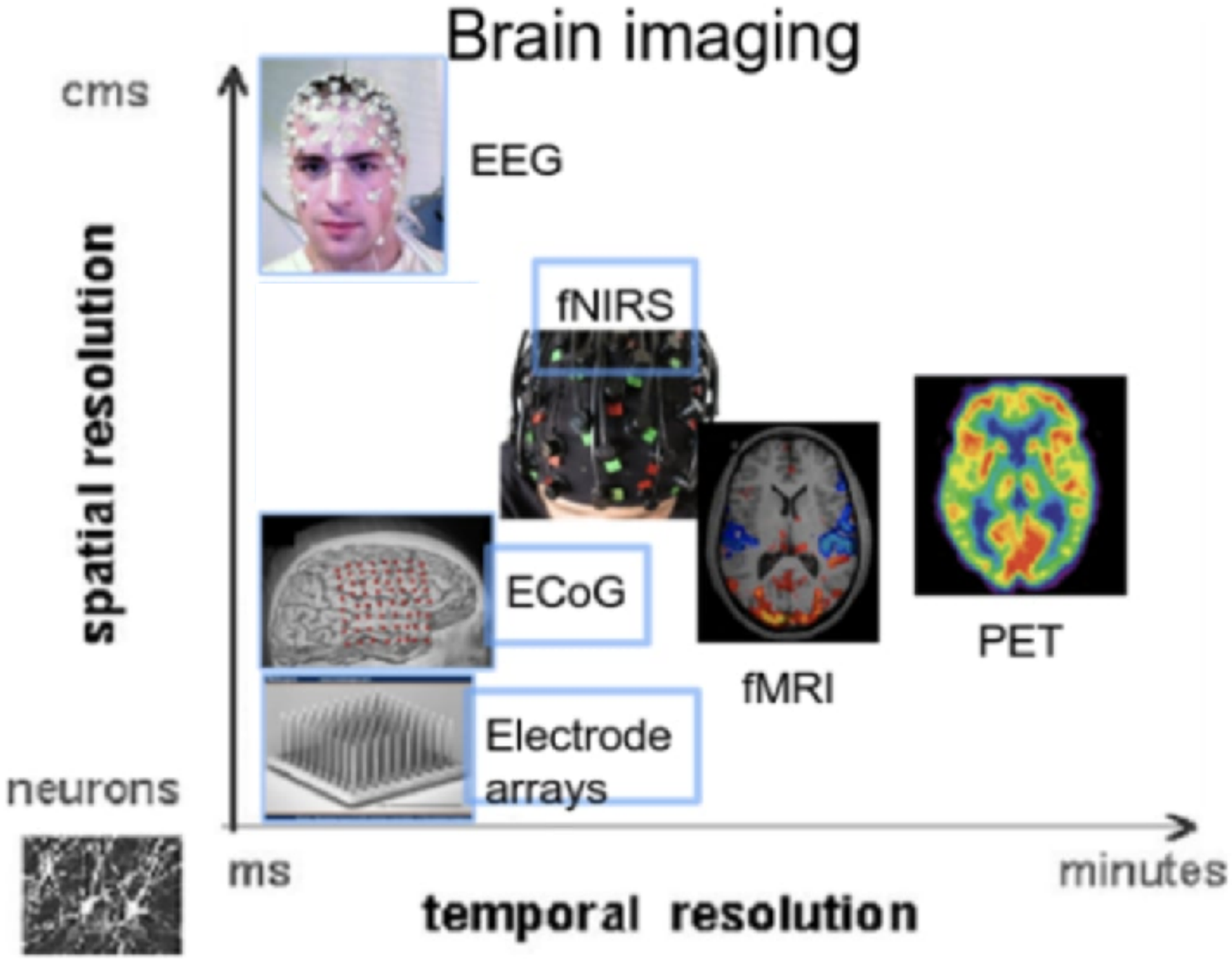


Image source: NeuroTechEdu

Future of brain imaging for mind decoding



https://youtu.be/Ecvv-EvOj8M?si=FUZ-jVaMya6uQf7l